



# Multiplexing Techniques

Time Division Multiplexing (TDM)

Frequency Division Multiplexing  
(FDM)

Compiled By:

Kartik R. Patel

EXTC Department

KJSSE, SVU



# Need of Multiplexing

**Bandwidth utilization is the wise use of available bandwidth to achieve specific goals.**

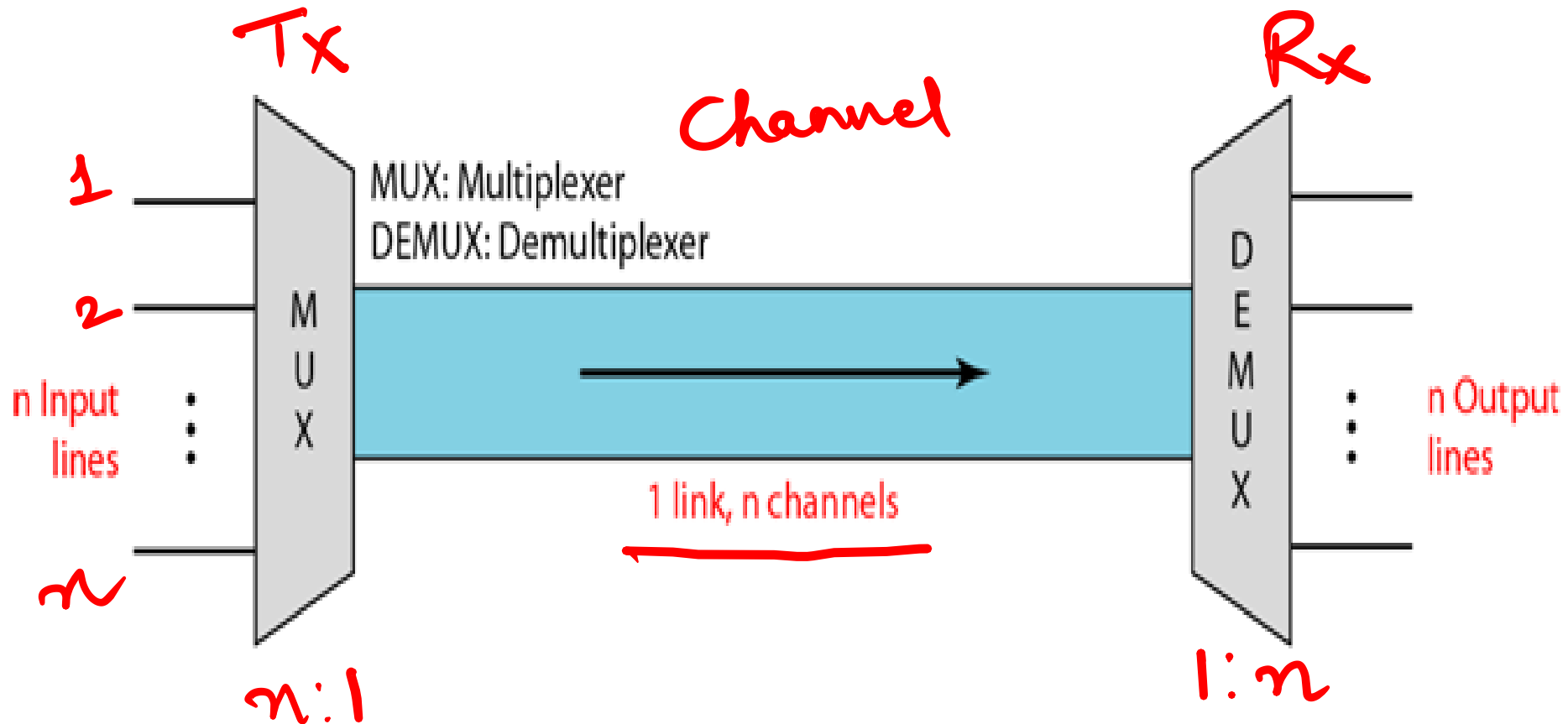
**Efficiency can be achieved by multiplexing; i.e., sharing of the bandwidth between multiple users.**

# MULTIPLEXING

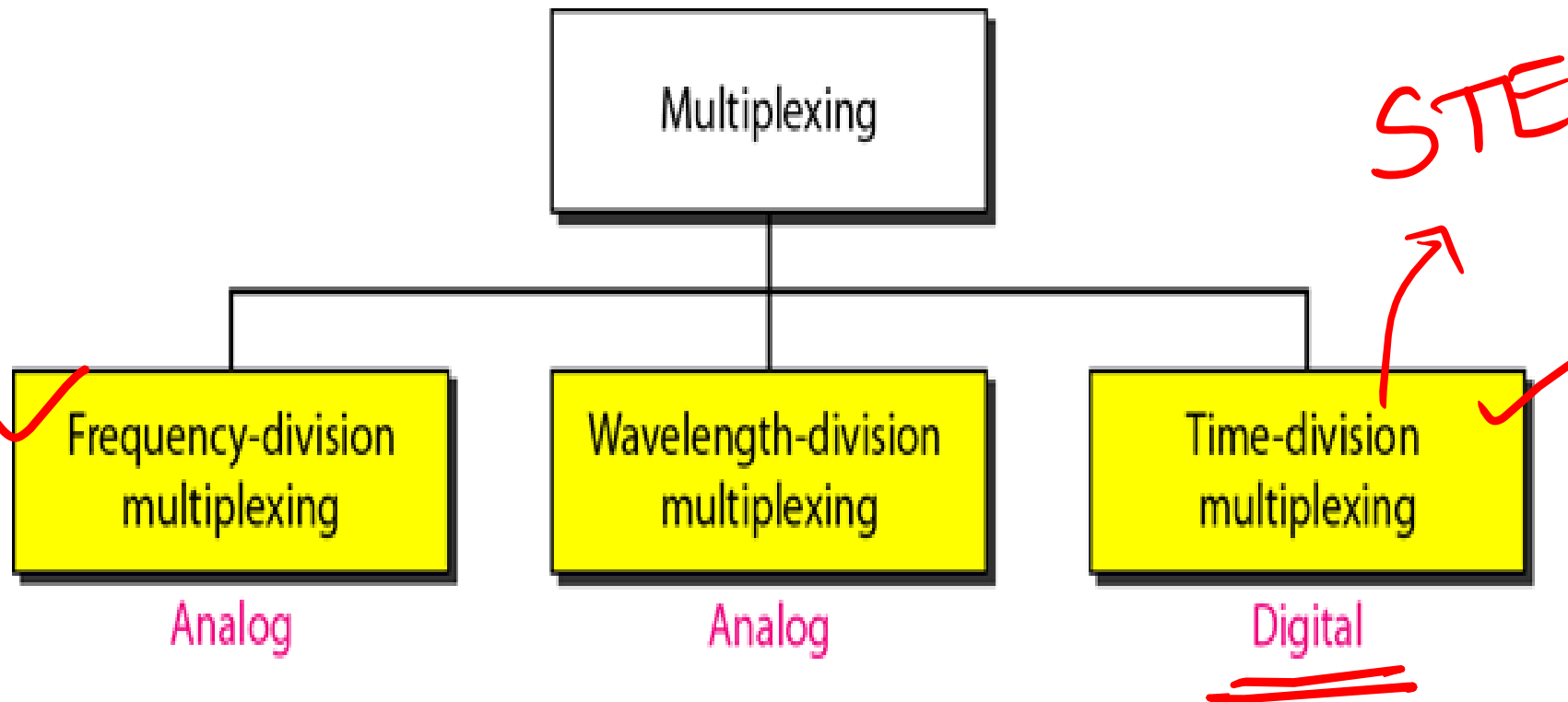
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*Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared. Multiplexing is the set of techniques that allows the (simultaneous) transmission of multiple signals across a single data link. As data and telecommunications use increases, so does traffic.*

# Multiplexing



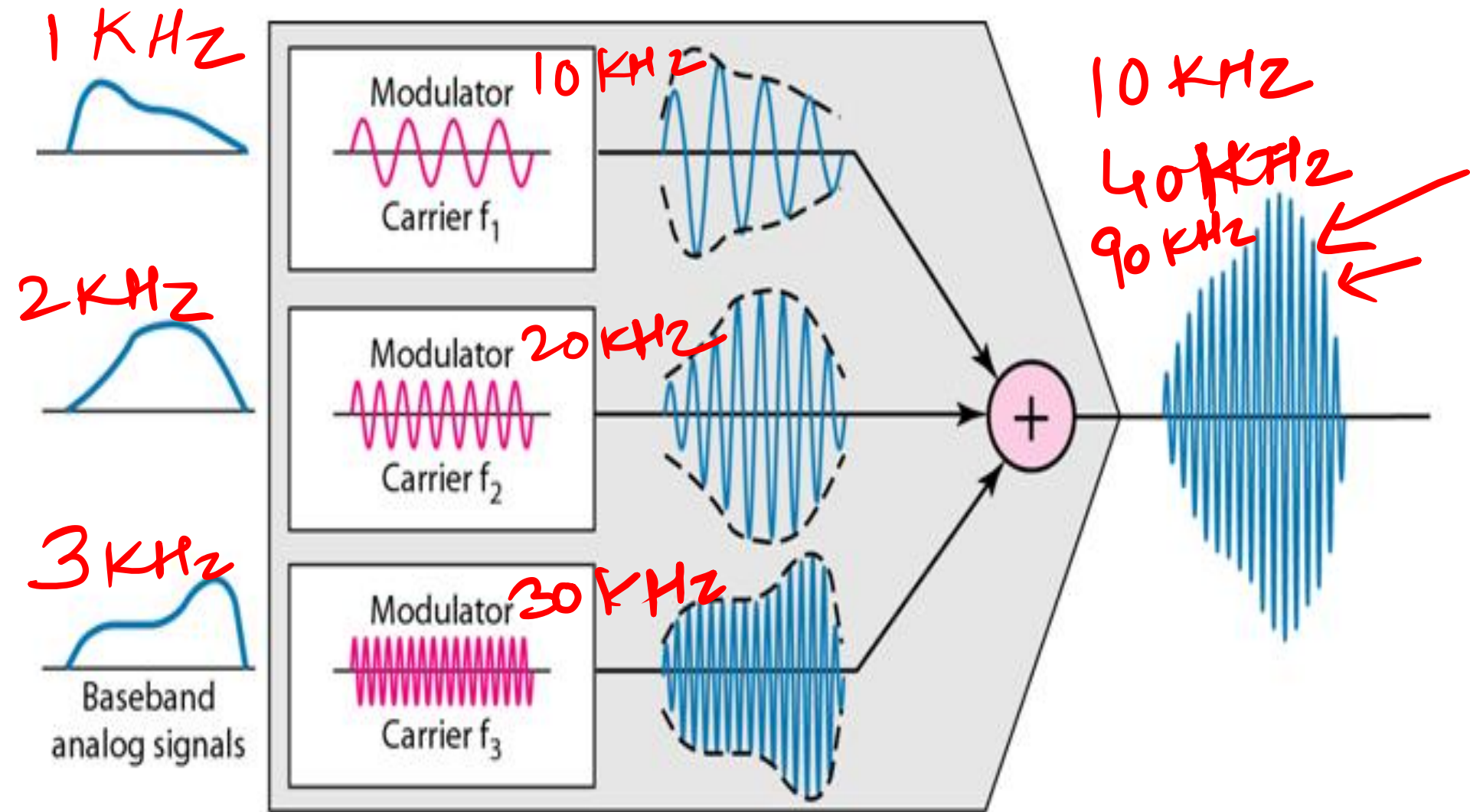
# Categories of Multiplexing



# Frequency Division Multiplexing



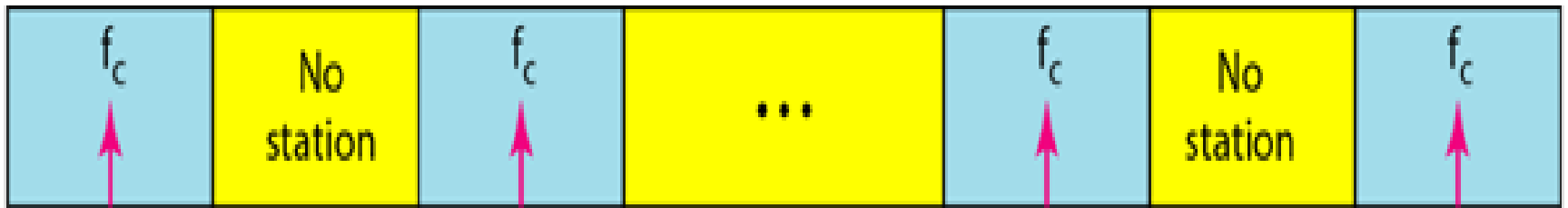
# FDM Process



FM

Example  
Ch-3

Ch-1 Ch-2 Ch-3 Ch-4



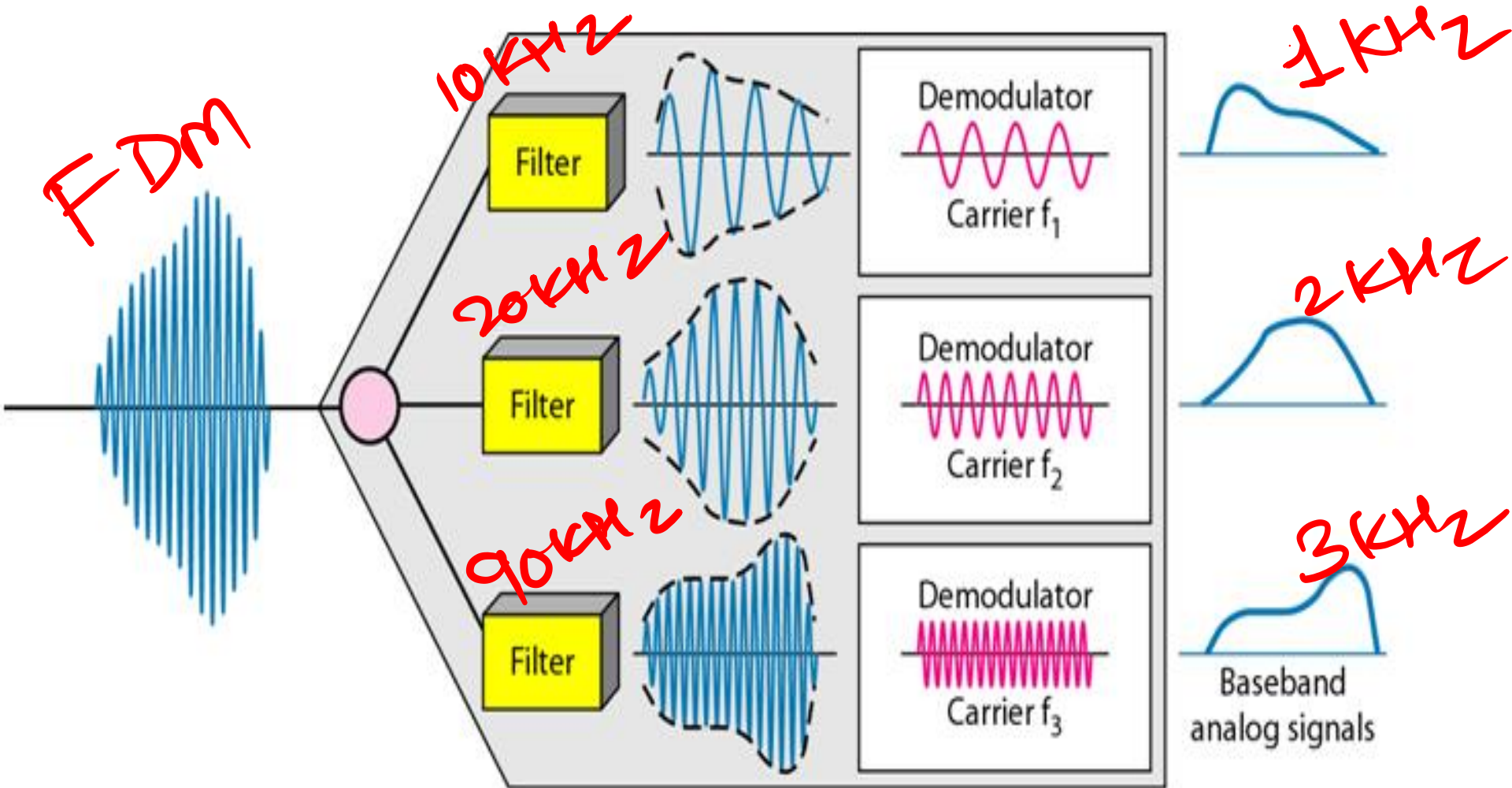
88  
MHz

108  
MHz

Guard  
bands



# FDM Demultiplexing



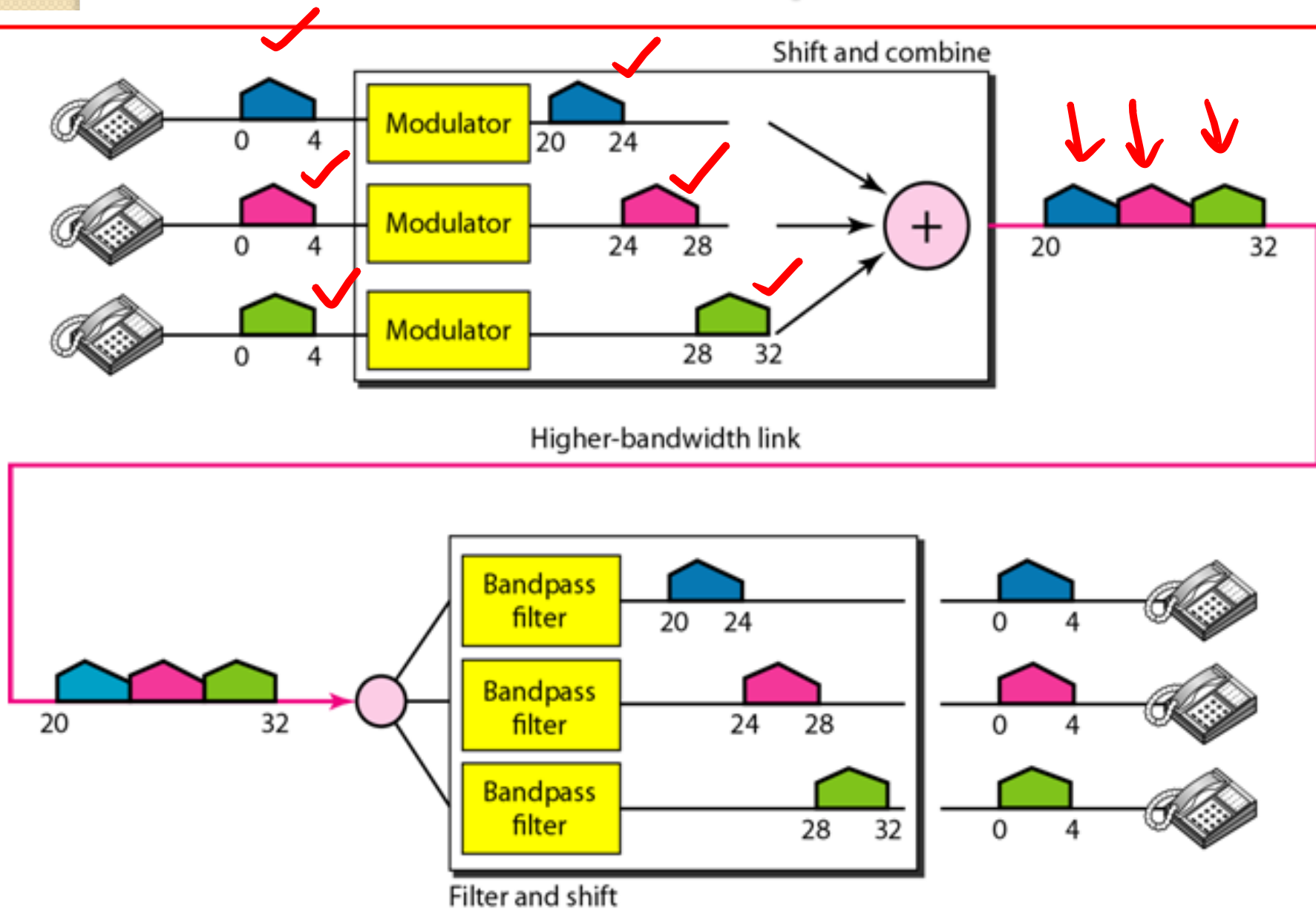
# FDM Example-I

*Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.*

## ***Solution***

*We shift (modulate) each of the three voice channels to a different bandwidth, as shown in Figure 6.6. We use the 20- to 24-kHz bandwidth for the first channel, the 24- to 28-kHz bandwidth for the second channel, and the 28- to 32-kHz bandwidth for the third one. Then we combine them as shown in Figure 6.6.*

# FDM Example

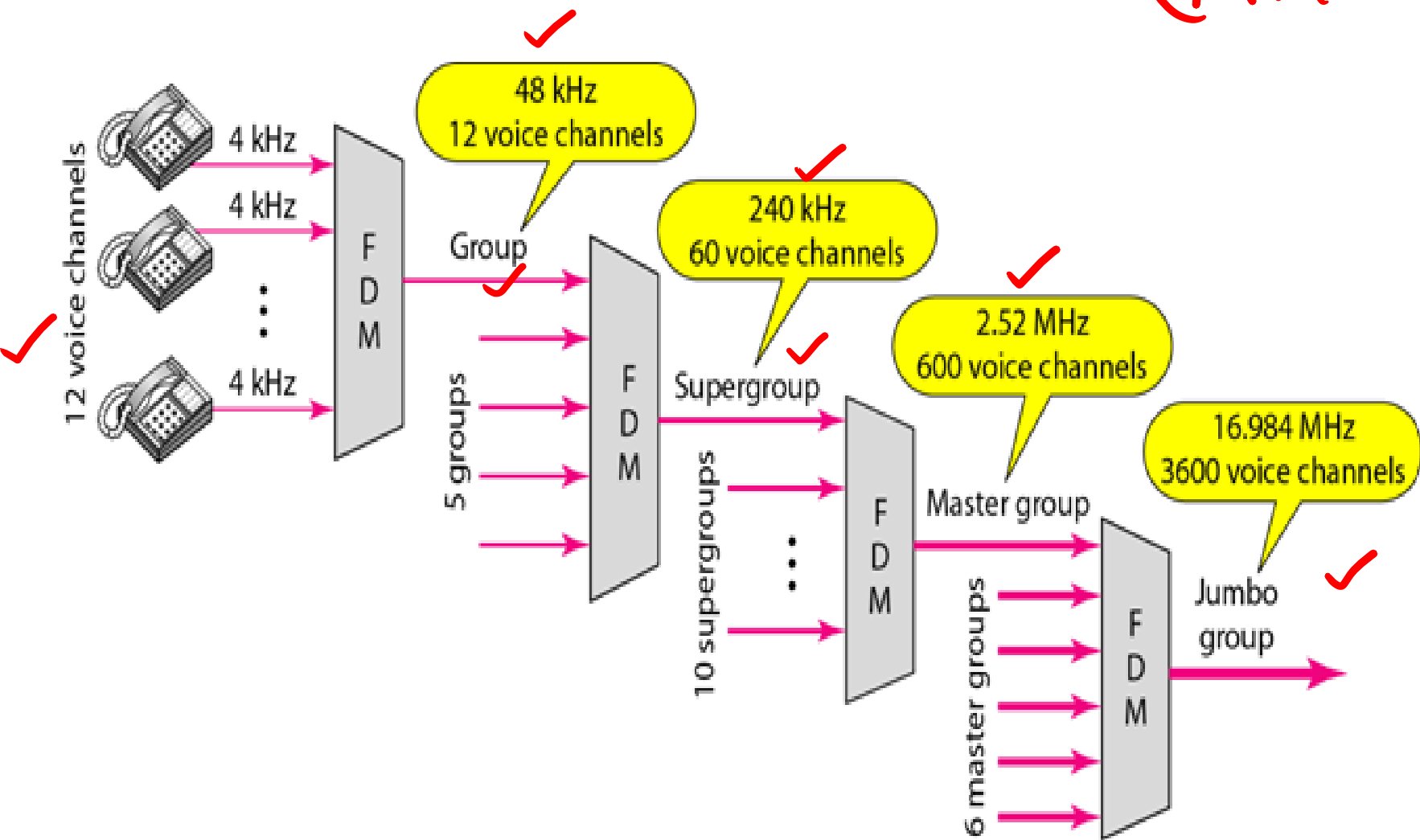


# FDM Example-2

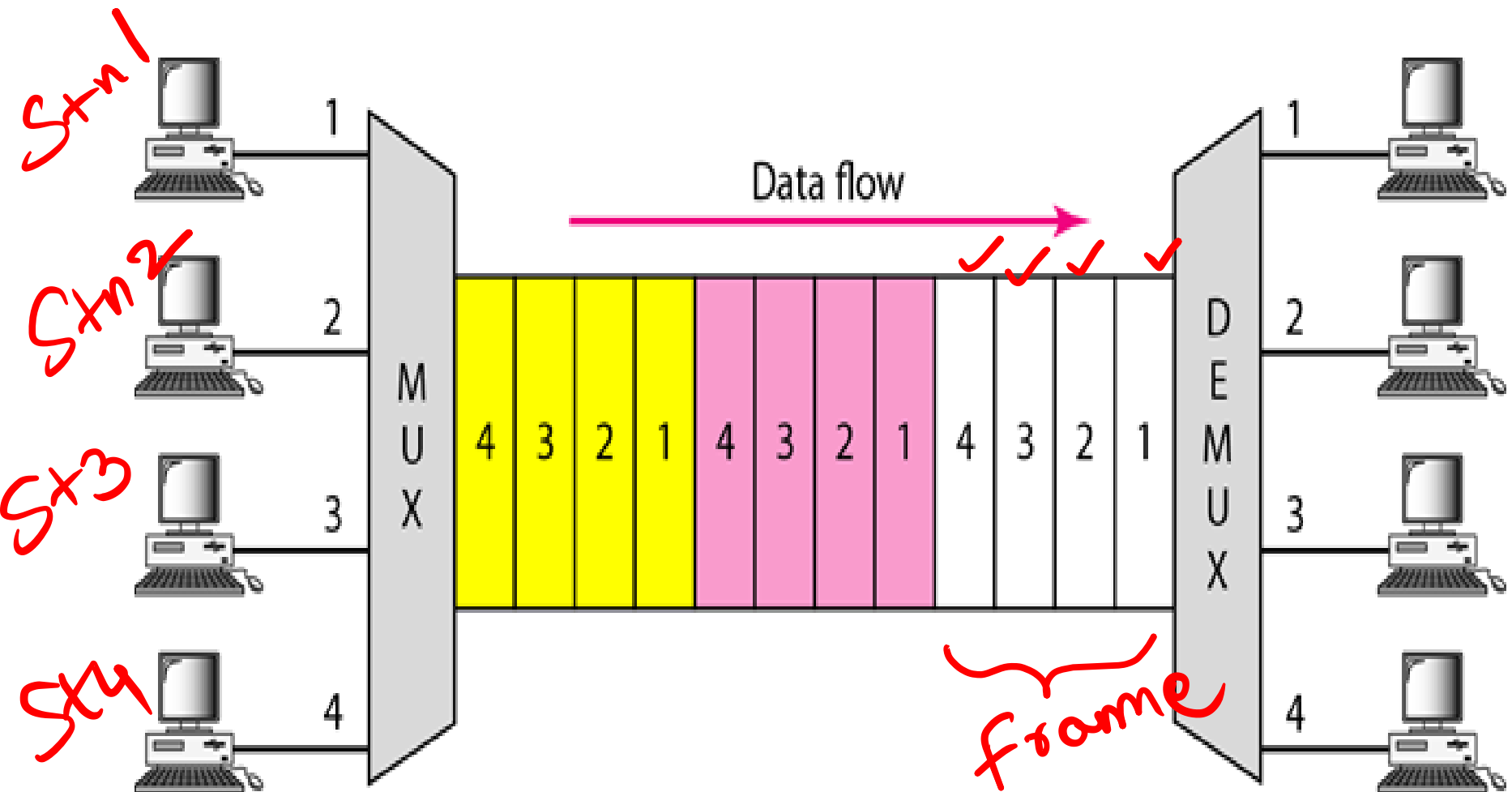
Example 02: Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?



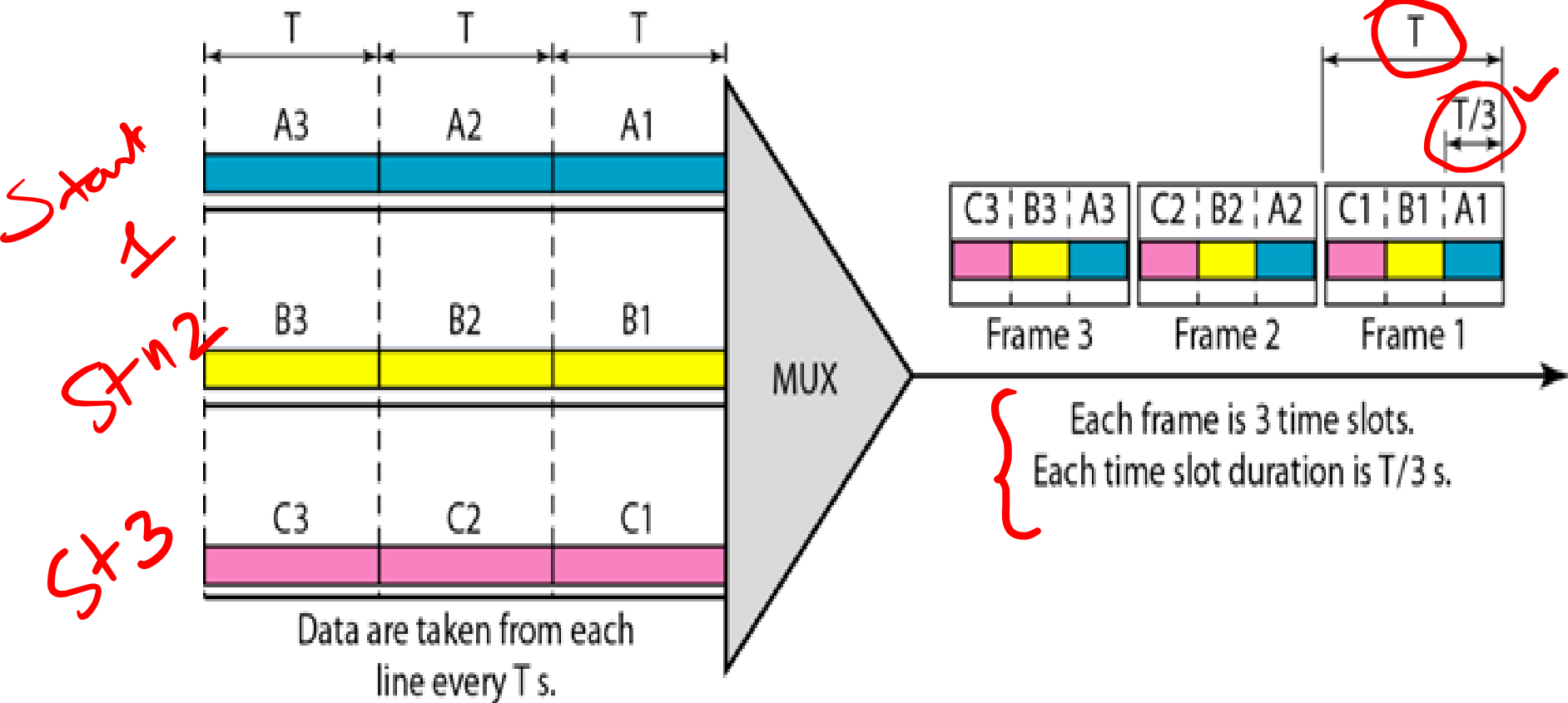
# Analog FDM Hierarchy (Application)



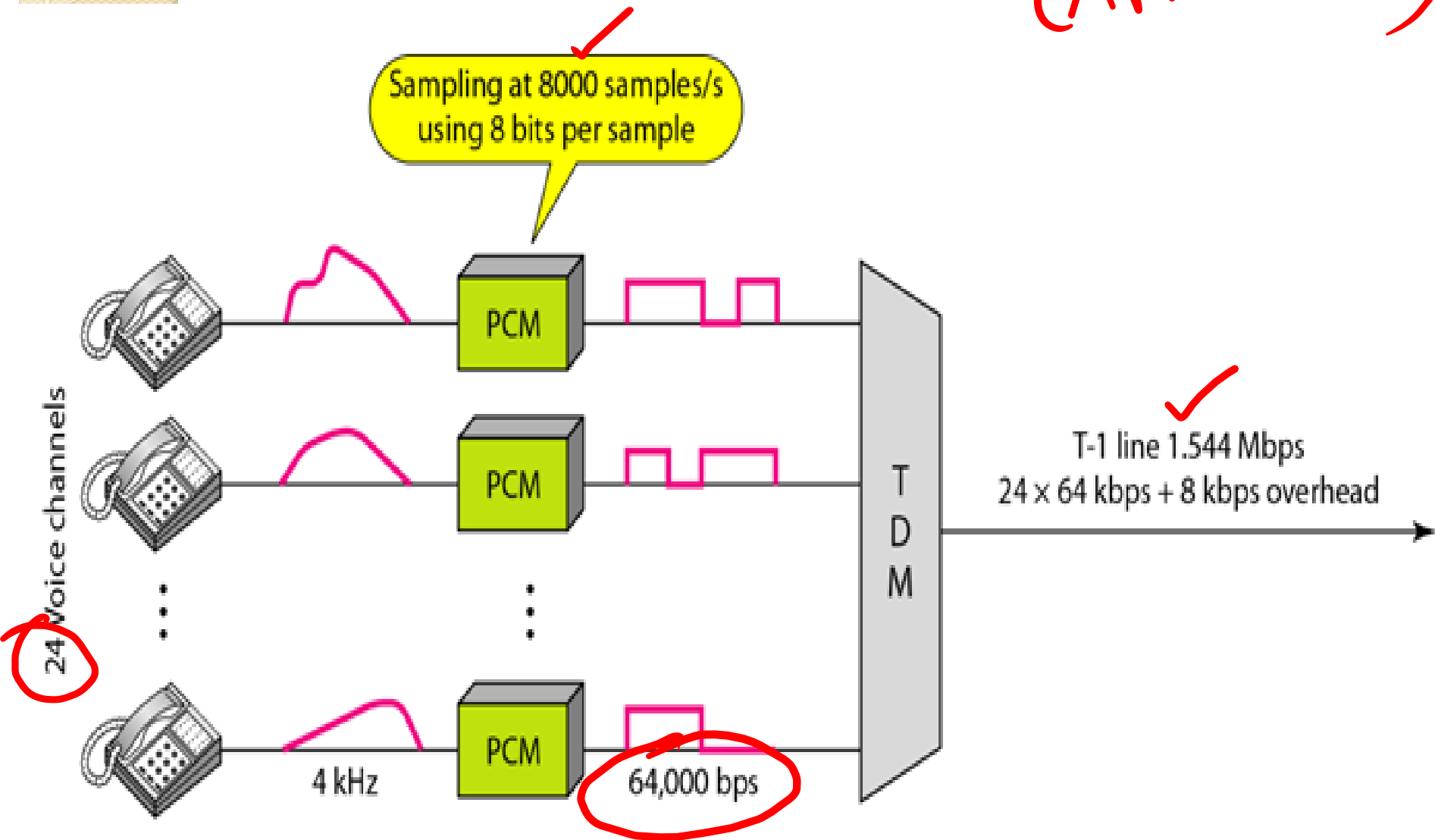
# Time Division Multiplexing (TDM)



# Synchronous TDM



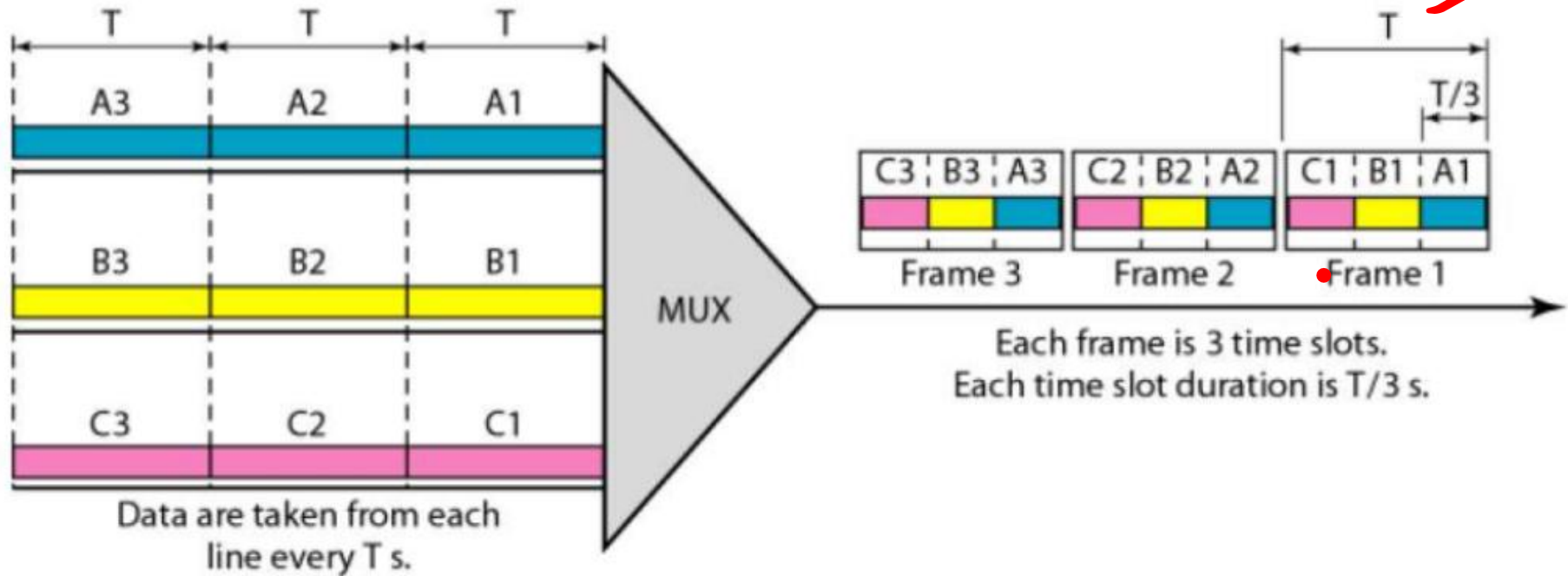
# TDM Example (Application)





# TDM Example-I

Example 03: In Figure, the data rate for each one of the 3 input connections is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of (a) each input slot, (b) each output slot, and (c) each frame?



# TDM Example-1

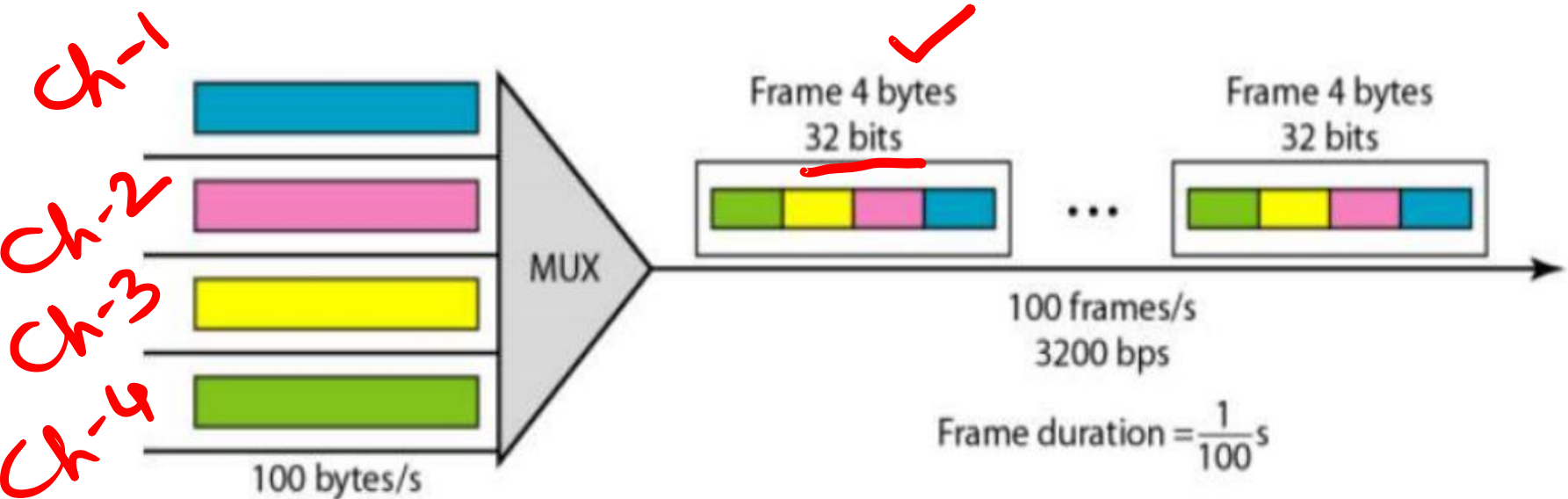
**Example 03:** In Figure, the data rate for each one of the 3 input connections is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of (a) each input slot, (b) each output slot, and (c) each frame?

- a. The data rate of each input connection is 1 kbps. This means that the bit duration is  $1/1000$  s or 1 ms. The duration of the input time slot is 1 ms (same as bit duration).
- b. The duration of each output time slot is one-third of the input time slot. This means that the duration of the output time slot is  $1/3$  ms.
- c. Each frame carries three output time slots. So the duration of a frame is  $3 \times 1/3$  ms, or 1 ms.

Note: The duration of a frame is the same as the duration of an input unit.

# TDM Example-2

**Example 04:** Four channels are multiplexed using TDM. If each channel sends 100 bytes /s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link.



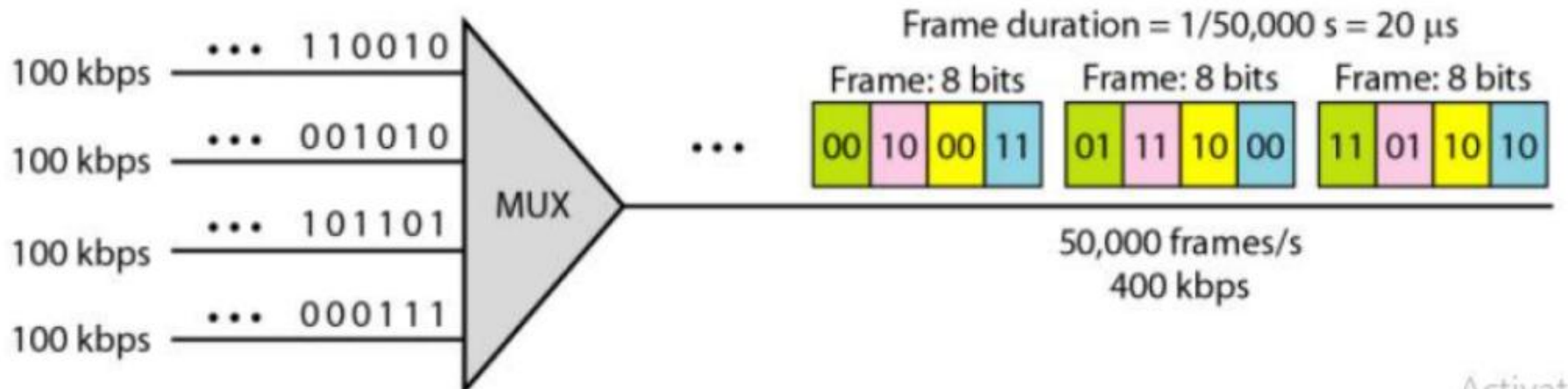
# TDM Example-2

## **Solution:**

The multiplexer is shown in Figure 6.16. Each frame carries 1 byte from each channel; the size of each frame, therefore, is 4 bytes, or 32 bits. Because each channel is sending 100 bytes/s and a frame carries 1 byte from each channel, the frame rate must be 100 frames per second. The bit rate is  $100 \times 32$ , or 3200 bps.

# TDM Example-3

**Example 05:** A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?



# TDM Example-3

## **Solution:**

Figure 6.17 shows the output (4x100kbps) for four arbitrary inputs. The link carries  $400\text{K}/(2 \times 4) = 50,000$   $2 \times 4 = 8$  bit frames per second. The frame duration is therefore  $1/50,000$  s or  $20\mu\text{s}$ . The bit duration on the output link is  $1/400,000$ s, or  $2.5\mu\text{s}$ .